



TECHNICAL REQUIREMENTS DOCUMENT

AI Sales & Operations Workflow — Build Specification for Subcontractor

Source Proposal Date: 1 September 2026

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Prepared for: [Subcontractor Name]*

1. Purpose & Scope

This document specifies the functional requirements for the AI-driven sales and operations workflow described in the client proposal “AI-Powered Sales & Operations Workflow” (1 September 2026), for build-out by a subcontractor on behalf of De AI Solutions Pte Ltd.

The subcontractor is responsible for building, integrating, and deploying the AI agent and its supporting automation so that it executes the workflow defined in Section 4 without manual intervention beyond the human checkpoints explicitly called out below.

Out of scope: on-the-ground flyer/banner marketing and the physical landscaping work on-site. These are manual, offline activities performed by the client and are shown in the workflow diagram only for context.

2. Actors & Roles

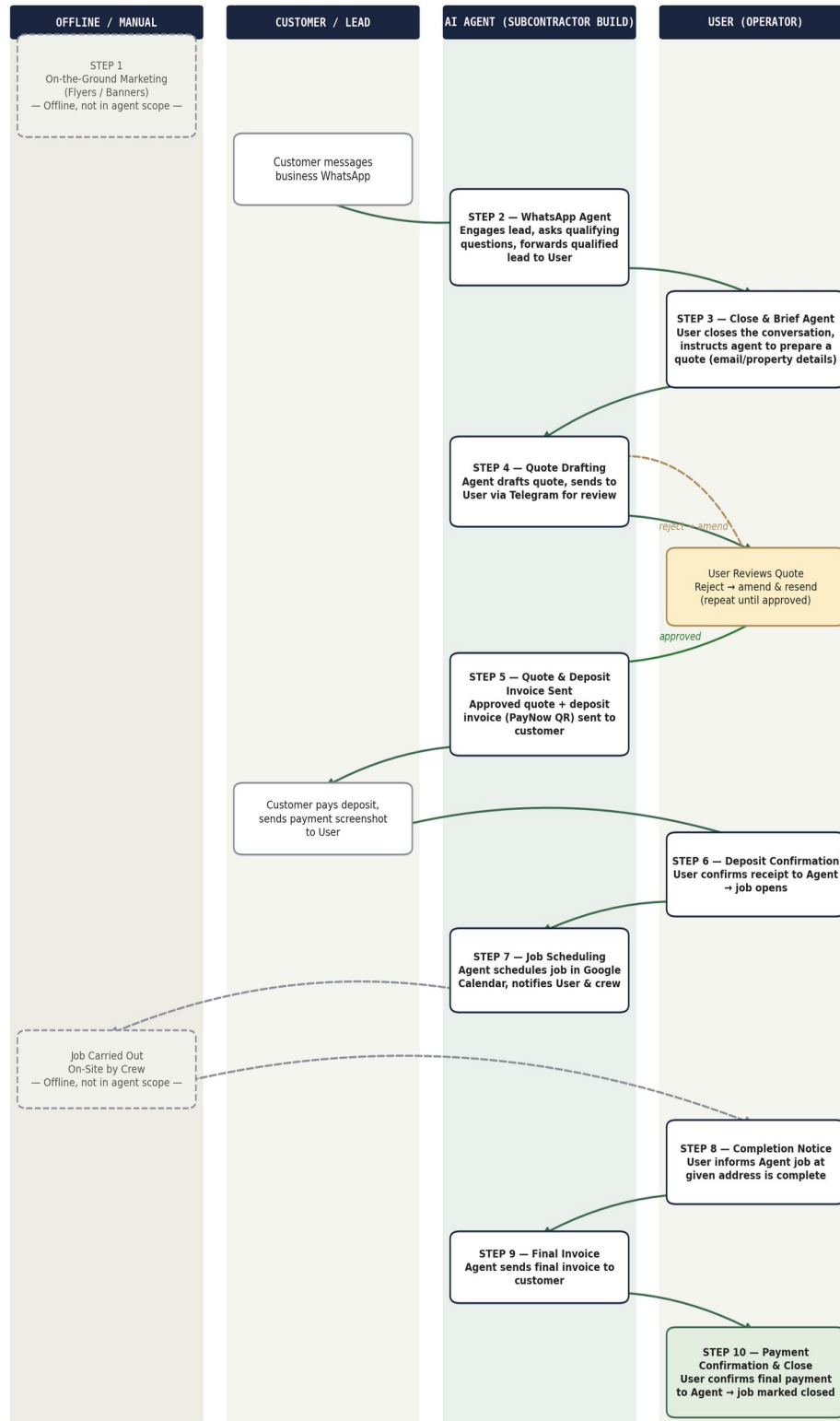
Actor	Role in the Workflow
Customer / Lead	Initiates contact via WhatsApp; receives quotes and invoices; pays deposit and final invoice.
AI Agent	The system to be built by the subcontractor. Qualifies leads, drafts quotes, sends invoices, schedules jobs, and manages all customer-facing and User-facing messaging.
User (Operator)	Business owner/operator. Reviews and approves quotes, confirms payments, and gives completion instructions via Telegram.

3. System Components

- Hosted AI agent runtime — 1 × Mac mini (client-provided hardware, hosted on De AI Solutions' side per original proposal).
- WhatsApp Business integration — for inbound lead qualification and outbound quotes/invoices.
- Telegram Bot integration — for User-facing review, approval, and completion notifications.
- Google Calendar integration — for job scheduling once a deposit is confirmed.
- Email capability — for quotes/invoices where the customer prefers email over WhatsApp.
- PayNow QR generation / display — embedded in deposit and final invoices.
- Automation/orchestration layer (e.g. Zapier + Telegram bot, or equivalent) connecting the above into a single pipeline.
- Lead & job data store — to track each lead/job through its lifecycle (see Section 5).

4. Workflow Diagram

The diagram below shows the full workflow across four lanes: offline/manual activity, the customer, the AI agent (subcontractor build), and User. Arrows show the sequence of hand-offs between actors.



5. Functional Requirements

Each requirement below corresponds to a numbered step in the workflow diagram. The subcontractor should treat each as an independent module with a clear trigger and output, so modules can be built, tested, and revised independently.

REQ-02 WhatsApp Lead Qualification

TRIGGER	Customer sends a message to the business WhatsApp number.
INPUTS	Inbound WhatsApp message text; customer phone number/profile.
AGENT ACTIONS	Agent engages the customer in a qualifying conversation (defined question set to be agreed with User — e.g. service type, property type, location, timeline). Agent determines whether the lead is qualified.
OUTPUT	Qualified lead record (contact details + qualification answers) forwarded to User via the agreed channel (Telegram).
NOTES	Question script and qualification criteria are not yet defined in the source proposal — to be confirmed with User before build.

REQ-03 Close & Brief the Agent

TRIGGER	User manually closes the qualifying conversation with the lead.
INPUTS	User's instruction to the agent (customer email or WhatsApp preference, property details).
AGENT ACTIONS	Agent receives User's brief and holds the details required to prepare a quote.
OUTPUT	Quote-drafting job created and queued for REQ-04.

REQ-04 Quote Drafting & Review

TRIGGER	Quote-drafting job created (from REQ-03), or a rejection/amendment instruction from User.
INPUTS	Property/customer details from REQ-03; pricing rules (to be supplied by User); previous quote version (on amendment).
AGENT ACTIONS	Agent drafts a quote and sends it to User via Telegram for review. On rejection, agent receives amendment instructions, revises the quote, and resends. This loop repeats until User approves.
OUTPUT	User-approved quote, ready for REQ-05.
NOTES	Agent must preserve version history of the quote through each amendment cycle for audit purposes.

REQ-05 Quote & Deposit Invoice Delivery

TRIGGER	User approves the quote (from REQ-04).
INPUTS	Approved quote; customer's preferred channel (email or WhatsApp); deposit calculation basis (% or fixed amount — to be confirmed by User per job).
AGENT ACTIONS	Agent generates a deposit invoice (with embedded PayNow QR) alongside the approved quote and sends both to the customer via their preferred channel.
OUTPUT	Quote + deposit invoice delivered to customer.

REQ-06 Deposit Confirmation & Job Opening

TRIGGER	Customer pays the deposit and sends a payment screenshot to User.
INPUTS	Payment screenshot (received by User, outside the agent); User's confirmation instruction to the agent.
AGENT ACTIONS	Agent receives User's confirmation that the deposit has been received and marks the job as open.
OUTPUT	Job status set to "open"; job queued for scheduling (REQ-07).
NOTES	The agent does not verify payment directly — confirmation is manual, via User. Flag this as a manual trust point in testing.

REQ-07 Job Scheduling

TRIGGER	Job marked "open" (from REQ-06).
INPUTS	Job details (address, scope, customer contact); User/crew calendar availability.
AGENT ACTIONS	Agent creates a calendar entry in Google Calendar for the job and notifies User and the crew of the upcoming job.
OUTPUT	Google Calendar event created; notification sent to User and crew.

REQ-08 Completion Notice

TRIGGER	User informs the agent (via Telegram) that the job at a given address is complete.
INPUTS	User's completion message, referencing the job/address.
AGENT ACTIONS	Agent matches the completion message to the correct open job record and triggers final invoice generation.
OUTPUT	Job status set to "completed, pending final invoice"; final invoice job queued (REQ-09).

REQ-09 Final Invoice

TRIGGER	Job marked completed (from REQ-08).
INPUTS	Original quote/job value; deposit already collected.
AGENT ACTIONS	Agent generates the final invoice (balance due after deposit) and sends it to the customer via their preferred channel.
OUTPUT	Final invoice delivered to customer.

REQ-10 Payment Confirmation & Close

TRIGGER	User confirms to the agent that final payment has been received.
INPUTS	User's confirmation instruction.
AGENT ACTIONS	Agent marks the job as closed and archives the full record (lead → quote → deposit → job → final invoice → payment).
OUTPUT	Job status set to "closed".

6. Data & Integration Requirements

- WhatsApp Business API (or equivalent) access, with permission to send template messages (quotes, invoices) outside the 24-hour session window where required.
- Telegram Bot API access for the User-facing review/approval/notification channel.
- Google Calendar API access (OAuth or service account) scoped to the relevant calendar(s).
- Email sending capability (SMTP or transactional email API) for customers who prefer email.
- PayNow QR code generation or a static QR asset embeddable in generated invoices.
- Persistent storage for lead, quote, job, and payment records, with status tracked through the full lifecycle (see REQ-02 through REQ-10).
- Audit trail / version history for quote amendments (REQ-04).

7. Non-Functional Requirements

- Hosting: runs on the client-provided Mac mini, with no additional on-site hardware required from the customer or User.
- Human-in-the-loop: no quote, invoice, or message may be sent to the customer without passing through the relevant User approval checkpoint (REQ-04, REQ-05).
- Reliability: agent should recover gracefully from a dropped WhatsApp/Telegram connection without losing an in-progress lead or job record.
- Data retention: lead and job records should be retained for at least the duration of the 3-month pilot for reporting purposes.

8. Deliverables

- Configured WhatsApp AI agent covering REQ-02, REQ-05, REQ-09.
- Configured Telegram bot covering REQ-03, REQ-04, REQ-06, REQ-08, REQ-10.
- Google Calendar integration covering REQ-07.
- End-to-end tested workflow from first WhatsApp contact through to job closure.
- Brief handover documentation covering configuration, credentials/access, and how to update qualifying questions or pricing rules.

9. Open Items for Subcontractor to Confirm

The following details are not fully specified in the source proposal and should be confirmed with De AI Solutions / User before or during build:

- Exact qualifying question set for REQ-02.
 - Pricing rules / rate card the agent should reference when drafting quotes (REQ-04).
 - Deposit calculation basis (percentage vs. fixed amount) and whether it varies by job type.
 - User's Telegram account/handle and preferred notification format.
 - Crew notification method for REQ-07 (e.g. shared calendar visibility vs. a separate message).
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Please direct any questions on this specification to Vignesh Mahendran, De AI Solutions Pte Ltd.